

RSM — Just-Math Chain (v9.3)

The explicit realization. The implicit skeleton is affine — straight/curved only, no metric; this chain renders it and states its own measure. Everything a rendering adds — coordinates, a sign convention, a metric, a completed circle — belongs to the rendering, and where a result depends on one it is said so in place. Not sealed.

Notation

- P_0 — the total paradox: the master pair (distinction/indistinction) at equality. Presenting faces: the empty face and the full face, sitting in native coordinates at nothing and without limit. Neither is a numeral. P_0 and P_n are one species at two scopes; the difference is what survives the collapse. An ontological-logical impossibility.
- $R_n, O_n, X_n, Y_n, G_n, B_n, P_n$ — frame; origin; the two modes (magnitudes, and slots rather than names); the gradient $X_n \cdot Y_n = 1_n$; the balance line $X_n = Y_n$; and $P_n = G_n \cap B_n$, at $(\sqrt{1_n}, \sqrt{1_n})$ in rendering coordinates.
- B_n **does not obtain at any point** (structural §5). Neither does P_n . The coordinates above are where the drawing puts them, not places anything is.
- σ_B — the mode-swap $X \leftrightarrow Y$; on G the inversion $x \mapsto 1_n/x$; in split coordinates it fixes a and negates b , fixed set $b = 0$. σ_B **is the structural involution**.
- v — componentwise negation, fixed point O_n . **v is explicit**. A sign is an indexed comparison, not a second obtainer.

The reciprocity requirement

1. **Co-obtaining (P2)**. Neither mode reaches a terminal while the other obtains.
2. **Exchange symmetry**. The law is symmetric under $X \leftrightarrow Y$ — the structural symmetry.
3. **Class selection**. Additive conservation fails twice: it admits $X = 0$ (a terminal as an attainable value) and it caps both modes. Multiplicative conservation does neither: $XY = 1_n$ admits no zero and no ceiling. Both of P_0 's faces are excluded here, at the gradient's two ends, by the law's own shape.
4. **Uniqueness up to reparametrization**. Any terminal-excluding symmetric law is a monotone reparametrization of the conserved product; the product form is the law in

the modes' native magnitudes, and the native coordinates are fixed by the impossibility itself — they are the ones in which P_0 's faces sit at *nothing* and *without limit*. The chart is anchored by the very thing it excludes. The native-magnitudes step is unsettled.

The frame, the unit, the floor

1_n is the constitutive magnitude and the minimal unit in one claim: any distinction beneath it is, by the Corollary, frame $n+1$ ("sub- 1_n " is unit-relative typing, not size — cross-frame magnitude comparison is undefined). The floor is the Definition's own boundary; no floor postulate exists. 0_n is a limit-approach, never a value.

The one law and its two slices

Over $\mathbb{C}$ with $z = a + bi$, one equation $Q(z) = 1_n$ carries two real slices:

- $Q_i = a^2 + b^2$ — the circle slice.
- $Q_j = a^2 - b^2$ — the gradient slice, with split unit j , $j^2 = +1$; in split coordinates $a = (X+Y)/2$, $b = (X-Y)/2$, the gradient $XY = 1_n$ is exactly $Q_j = 1_n$.

For magnitudes X , Y the sum a is positive, and **the mode-expressible domain is the branch $a \geq \sqrt{1_n}$** — the Amplitude Floor. The branch is what the modes express; the remainder of the circle is the chart's continuation of it, and $b = 0$ on that branch is B_n , which does not obtain.

The circle slice as a complete object requires $a < 0$ and is — a legitimate indexed continuation. Its content where it is mode-expressible stands.

Scale and lineage. Within a frame, scale is motion along G : the split action $e^{\{j\phi\}}$ — hyperbolic rotation — is the orbit of rescalings $(\lambda X, Y/\lambda)$, product invariant. The family parameter $c = XY$ names the lineage direction along B : $c \rightarrow 0$ child-ward, $c \rightarrow \infty$ parent-ward, root-capped; a lineage coordinate, never a global count. The lineage direction is not scale.

The two non-terminations exchange, pivoting on P_n : the inversion $x \mapsto 1_n/x$ — σ_B on G — swaps the fine center-ward direction and the vast extension. Both non-terminations are the gradient's own ends, within the frame. This is the chain's cleanest computation of the structural involution.

The Archimedean demonstration

The explicit register's cleanest exhibition of the framework's central facts is twenty-three centuries old. Archimedes bounded π between inscribed and circumscribed polygons: the inner perimeters climbing, the outer falling, the value trapped between two sequences that

tighten forever and meet at no finite stage. Read structurally:

- **A limit approached from both sides and occupied by neither.** The inner and outer sequences are the two-sided approach to a locus no member of either sequence attains — the same species of fact as the gradient's approach to the balance line, and the explicit register's way of exhibiting it.
- **Each polygon is a frame.** The hexagon's ratio is exactly 3 — correct for a hexagon. The 96-gon's is 3.1416 — correct for a 96-gon. Each refinement is accurate at its own resolution and is not the thing it bounds. Indexed statements of one invariant: 可名 of a single 常名.
- **The non-terminating expansion is the engine's signature in a number.** Each further digit is the pursuit of exactness continuing below the previous digit's floor. The expansion does not terminate, and not from weakness of method: what it measures is the relation between straight and curved, and those are the skeleton's own two categories — the affine layer knows exactly this distinction and no metric. That the ratio between them can be bounded to any precision and completed at none is deferred exactness, rendered numerically.

Placement, stated: π 's numeric content — lengths, the value itself — enters with the measure, downstream, like perpendicularity. What is register-clean is the shape of the situation: minimal closure by straight elements takes three sides (an affine fact); the two-sided bound; the refinement without end.

Theorems

Disconnection. B_n does not obtain, so the X-dominant and Y-dominant portions of the mode-expressible branch have no path between them: any path joining them meets $b = 0$. The two portions are related — each is the other under σ_B — and not connected. This holds in the modes' own magnitudes, with no sign convention and no second component.

Revolution. The relation between the two portions is carried by revolution of G_n about B_n . A half-turn about B realizes σ_B as continuous motion, so the two portions exchange with nothing passing through $b = 0$. Revolution requires a direction the plane does not have; how many such directions the structure requires is open. The rotational freedom about B is the ring of orientations, and one selection is one branch.

The parturition map. The recursion is that $O_{(n+1)}$ reads P_n and $yAxis_{(n+1)}$ reads B_n , as constitution and not site-selection. $w = z^2$ realizes it, 2-to-1 with the two preimages differing by v . The two preimages are not two things bound into one; they are **one branch compared from the child's side** — indexed comparison performed across a frame boundary. This is how a sign can be measured while being explicit.

T2 — crossing — computation checked; the claim belongs to the rendering. Any closed spanning path crosses the null cone $Q_j = 0$; at the sign-image of P_n the gradient form reads -1_n . Spanning presupposes two components to span.

The two returns — computation checked; the claim belongs to the rendering. Along the circle, $Q_j = 1_n \cdot \cos 2\theta$: the invariant returns at π , the position at 2π . A double cover — a fact about the circle slice. Euler's identity is this fact compressed: $e^{i\pi} = -1$ says a half-turn lands on the sign-image — the most celebrated equation in mathematics as the signed chart describing its own return structure. A remark about the rendering, entered as one.

$i^2 = -1$, two routes. Elimination over the five candidate closures, and the antipodal-binding argument. Results about the signed continuation.

The v-centroid. Any sign-symmetric configuration sums to the center.

Complementarity floor — belongs to the rendering; the winding-measure premise is unsettled. Over admissible closed spanning structures, $\sigma_a \cdot \sigma_b \geq 1_n/2$, saturated by the minimal traversal.

The measure

The measure is definitional-by-register: Q_j is the circle slice's own quadratic form, and distances within the rendering are read from it. Whether quadraticity is forced rather than adopted is open — a non-quadratic candidate satisfying the coarse constraints exists and fails only the parallelogram condition. Until quadraticity is forced, "orthogonal" is a rendering fact: the skeleton says *independent*, and perpendicularity enters with the measure, downstream. The register the measure is defined on is itself explicit as a complete object.

The engine (realized)

Continuity entails B, and G meets it at P_n . Denial A (origin): one constitutive pair per frame; n has no spare pair to grant the locus. Denial B (exactness): at exact equality nothing distinguishes the modes, so the condition does not obtain. Satisfaction as organizing-locus: in-frame, circulation about a locus nothing occupies; trans-frame, generation — **the child is the pursuit of exactness continuing below the floor**, its origin the approached locus read from below, its axis the parent's balance line, realized by the parturition map. Recursion is exactness, permanently deferred.

No address. P_n 's coordinates are where the rendering puts it; nothing is sited there. The standing residue of sub-unit imbalance entails the child. The incident-pair deficit $(1 - 1_n/x_0^2)$, rendered as perpendicularity once the measure is in force) is corroborating, never load-bearing.

Anchoring. Every frame beneath n 's unit and in relation to n is parent-seeded: "beneath 1_n " is defined only relative to the parent's unit, so a sub-unit frame is by definition one the parent seeds; a free-floating sub-unit frame is a contradiction in the typing.

What this chain does not owe

No exponent, no dimensionful constant, no scaling law. The implicit register fixes structure and fixes no parameters; a branching law with exponent two and one with exponent three instantiate the same structure. Quantities enter only with an instance, and instances are the explicit register's business.

Open items

1. **The open closures** — the native-magnitudes step; the measure's definitional typing; the anchoring argument.
 2. **The cross-frame unit relation** $1_n \leftrightarrow 1_{(n+1)}$ — whether any quantitative relation accompanies the constitutive share. An explicit-register question.
 3. **Dimension** — revolution about B requires at least one direction beyond the plane, and the full argument runs further: with no orientation privileged, the complete set of arcs over the center sweeps a sphere, and a sphere requires at least three dimensions while requiring no fourth. "At least three" therefore has a route; that nothing *requires* a fourth is not yet that nothing can use one, so "exactly three" remains open (two rival derivations, unadjudicated).
 4. **Cross-frame expression** — how a parent states any fact about a child's interior.
 5. **Definition T (traversal)** — the formal object realizing ordered passage. Traversal as previously stated runs in the signed chart and needs restating in the mode-expressible domain, where the disconnection theorem now lives.
 6. **The tree-question** — sibling structure and commensurability (shared ancestry as the only common register).
 7. **Branching beyond P_n** — whether any relation other than the constitutive pair can be inexact in the constituting way. Candidate, unruléd: every conjugate equality within the frame reduces to $x = 1_n/x$, and the other points of B belong to other frames.
 8. **Quantitative complementarity** — the winding-measure premise; the non-star-shaped extension. A question about the rendering.
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v9.3 — the chains state the framework as currently known; the editorial record holds how it got here.